Starters: From Scratch, Dried, and Revived

The 14-day method, what is happening in the jar, and a backup that comes back from a pinch

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

1. **The 14-day method**: Full Proof Baking's schedule in grams, and her rules
2. **What is happening in the jar**: the early rise is bacteria; the day-5 lull is the hand-over
3. **Video, literature and this kitchen**: where the explanations differ; warm, not hot
4. **Drying a backup and reviving it**: a pinch, not a spoonful
5. **Open tests**: seven experiments the note leaves open

Flags carried over from the notes: **hypothesis** **not yet tested** **not peer-reviewed** **loose analogue**.
Results from this kitchen are marked **own result**; everything else is cited at the foot of the slide.

Source: notes/starter-from-scratch.md and notes/starter-cheatsheet.md

The 14-day method

What you need: a 30% rye mix, a rubber band and 24°C

- **Flour mix, 30% rye:** 350 g bread flour + 150 g dark rye, whisked together. Every feed until day 12 uses it
- **A straight-sided jar** with a rubber band at the level of each fresh feed; a fresh jar on day 2, and whenever it gets messy
- **A scale reading to 0.1 g**, tared before each ingredient: the feeds go down to 5 g
- **24°C** (75°F), "warm room temperature", for all 14 days
- **Cheesecloth** and a rubber band until day 8, then a loosely fitted lid

Ratios are **starter : water : flour by weight**, so every feed is at 100% hydration.

In the kitchen: Rye for its nutrients and enzymes, bread flour for strength. Plain flour also works for the base.

Source: Kristen, Full Proof Baking (2019), from her on-screen title cards and narration

Two weeks in grams: 1:1:1 for a week, then the ratio climbs

Day	Morning, g	Evening, g	What you'll see
1	—	0 / 30 / 30	Mix only, rest 24 h
2	—	Stir, clean jar	Dry skin on top: fine
3	—	20 / 20 / 20	Risen: bacteria, not yeast yet
4	20 / 20 / 20	16 / 16 / 16	3–4×, runny, cheesy smell
5	**15 / 15 / 15**	**16 / 16 / 16**	**Almost no rise, vomit smell: normal**
6–7	15 / 15 / 15	15 / 15 / 15	1.2–2×, milder, first sour notes
8	15 / 15 / 15	15 / 15 / 15	~2×. Save discard, loose lid
9	15 / 15 / 15	8 / 16 / 16	2.5×
10	8 / 16 / 16	5 / 15 / 15	2.5×, falls before the feed
11	5 / 20 / 20	5 / 25 / 25	2.5×
12	5 / 25 / 25	5 / 20 / 20	Evening: switch to 10% rye mix
13	5 / 25 / 25	5 / 25 / 25	2.5×
14	8 / 16 / 16 ×2	10 / 50 / 50	Feeds 7 h apart; three a day now

Source: Full Proof Baking (2019), as tabulated in the cheat sheet. Feeds in grams of starter / water / flour: two a day from day 4, three from day 14

Feed at peak; it is done when it triples in 4–6 h at 27°C

Feeding

- Fallen before the feed? Next feed one ratio up: 1:2:2 → 1:3:3 → 1:4:4 → 1:5:5
- Fewer than three feeds a day: higher ratios, up to 1:10:10
- Day 12: 10% rye mix = leftover 30% mix + twice its weight in bread flour, for a “higher peak, hold peak longer”
- **Discard** if pink, orange or fuzzy

Done

- Triples in 4–6 h on 1:2:2 at **27°C**, not room temperature
- No 40°C+ builds until done. Fridge: 2–4 weeks unfed

Video, day 14	2× at	Peak
1:5:5	5 h	2.9× at 7.5 h
1:2:2	4 h	3× at 7 h

Not there yet: “still pretty young”

In the kitchen: Her closing card: “Feed at peak! Monitor your starter! Keep it warm!”

Source: Full Proof Baking (2019); the day-14 time-lapse, in an oven with the light on at 27°C (80°F)

What is happening in the jar

The first two weeks are a succession of microbes

	Days 1–4 the first wave	Day 5 the lull	Days 6–9 the hand-over	Days 9–14 building strength
In the jar	*Leuconostoc*, *Lactococcus*, *Enterococcus*, *Enterobacter* from the flour	Acid stops the first wave; the sourdough organisms are not yet in numbers	*Lactobacillus*, *Pediococcus*, *Weissella* take over; yeast the day after pH 3.5	Community largely settled; what remains is matching food to appetite
She saw	3–4× rise, unpleasant smells	"No significant growth!", the worst smell	1.2× → 2×; "bit of sour" on day 7	Steady 2.5×; the ratio climbs

In the kitchen: The early rise that dies and the smell put people off, but they are an ordinary, well-documented succession.

Source: The two lab studies, both refreshed daily: Van der Meulen et al. (2007) at 30°C; Ercolini et al. (2013) 5 h a day at 25°C, 10°C between. Also Wink (2008a, b); Full Proof Baking (2019)

Days 1–4: the early rise is bacteria, not yeast

- Flour brings mostly *Proteobacteria*: *Pantoea*, *Pseudomonas*, *Enterobacter*. Within a day all but the *Enterobacteriaceae* are suppressed and lactic acid bacteria (LAB) take over
- The first LAB are not sourdough organisms: *Leuconostoc*, *Lactococcus*, *Enterococcus*. At 30°C they fall below detection within a day or two
- Rye is the exception: it arrives with *Weissella* already dominant
- Wink had a day-two starter analysed by two labs. The gas came from *Leuconostoc citreum*, with a little *Enterobacter cloacae*; neither lab found lactobacilli or yeast **not peer-reviewed**
- She links *Enterobacter*, *Aerococcus* and *Leuconostoc* to the bad smell

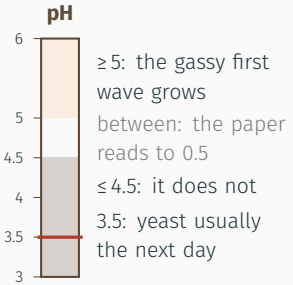
In the kitchen: The 3–4× rise and the smell on day 4 are this first wave. Keep to the schedule.

Source: Ercolini et al. (2013); Van der Meulen et al. (2007); Wink (2008b), not peer-reviewed

Day 5: the lull is acid shutting down the first wave

- As acid builds, the first wave stops before the sourdough organisms have numbers: “No significant growth!”
- Wink’s pH paper: gassy bacteria grew at pH ≥ 5, not ≤ 4.5; yeast usually came the day after pH 3.5

  careful amateur work; no replicate counts
- **hypothesis** (hers): *L. citreum*’s bacteriocins hold back the lactobacilli, so a bigger early rise means a longer lull
- Lab data agree: at 30°C, pH below 4.0 after day 2; yeast at 10^7 CFU/g after 4–5 days (one dough in four: 8)



In the kitchen: Almost no rise and a vomit smell on day 5 is normal. A pH strip at each feed, days 3–7, tests Wink’s thresholds.

Source: Wink (2008a, b), not peer-reviewed; Van der Meulen et al. (2007)

Days 6–9: the sourdough bacteria take over, then little changes

Sourdough-specific LAB, such as *Lactobacillus*, *Pediococcus* and *Weissella*, replace the first wave.

Where	Conditions	When it settles
Van der Meulen et al. (2007)	30°C, refreshed daily	Well-adapted strains dominate from about day 5
Ercolini et al. (2013)	25°C, 5 h a day	Mature in 5–7 days; lactobacilli prominent by day 10
The video	24°C	“Bit of sour” on day 7: about a day behind 30°C

In the kitchen: The first sour smell, with the rise back to 2×, is the hand-over: sourdough bacteria replacing the first wave.

Source: Van der Meulen et al. (2007); Ercolini et al. (2013); Full Proof Baking (2019)

Days 9–14: raising the ratio is about timing, not species

- By day 9 the community has largely settled. What remains is matching food to appetite
- A 1:2:2 feed that peaks and collapses within 12 h ran out of food well before the next feed, and then sits in its own acid
- A higher ratio dilutes the starter, so it takes longer to peak, and dilutes the acid it carried over: the same lever as chapter 11's recovery at 1:10:10
- "Feed at peak" is **a practitioner's rule, not a measured result**
- The plausible mechanism is chapter 11's compounding stressors: past peak the food is spent and acid and ethanol are at their highest, so a starter always fed after collapse is always fed at its most stressed

In the kitchen: If it peaks well before the next feed is due, raise the ratio. That is the whole of days 9–13.

Source: Full Proof Baking (2019); chapter 11, *Compounding Stressors* and *Recovery Protocol*

Video, literature and this kitchen

The method works; three of its explanations do not

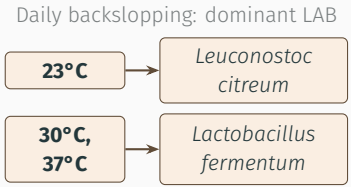
	The video says	The evidence says
1	It “picks up the wild bacteria and yeasts from the air”	**The flour, then the baker’s hands.** Same recipe and flour on two continents: much of each community traced to the flour, the differences to the bakers’ hands. In bakeries over a year, flour drove the LAB; the bakery’s own microbes played a minor role. It happens in a covered jar too
2	Day 3: “We’ve caught some wild yeast!”	In a two-day-old starter the rise is bacterial. Her narration says “not the correct balance” at once: only the label is wrong
3	Day 8: a loose lid, as “this is a live culture that requires air”	Yeast and LAB ferment without oxygen. The lid is loose so CO_2 can escape, not so air can get in

In the kitchen: The cheesecloth does no harm, but it is not what inoculates the jar: the flour is.

Source: Reese et al. (2020); Minervini et al. (2018); Wink (2008a), not peer-reviewed

Start warm, not hot: 24°C or 30°C picks a different culture

- New starters are small and unsettled, like chapter 11's depleted culture: 22–25°C, no selection pressure
- Warm is fine: her target is 27°C; Van der Meulen et al. (2007) had a stable culture in a week at 30°C
- *L. citreum* is behind Wink's false start: at 24°C the first wave may never fully leave
- **hypothesis** begun near 30°C, a starter may suit a warm-kept regime better (open tests 1–2)



Van der Meulen et al. (2011), via chapter 09

In the kitchen: No hot builds until it passes the tripling test; then keep the mother at room temperature and build hot from it.

Source: Van der Meulen et al. (2007, 2011); Wink (2008b), not peer-reviewed; chapters 08, 09, 11

Tap water and ordinary flour will do; juice is optional

Water

- Bottled water is usually about chlorine. Danish tap water is groundwater, normally unchlorinated: tap water should do

Flour

- Any bread flour for the base; the Manitoba is more than strong enough
- Dark wholemeal rye (*fuldkornsrugmel*): the whole grain is the point, as when chapter 11 adds wholemeal in recovery

A shortcut through the lull

- Wink: pineapple juice (or orange juice, or apple cider) for water on days 1–3 acidifies the mix below what the gas producers tolerate
- Yeast on or before day 4, against about day 7 with water
- It “doesn’t change the end result”; never necessary, “only patience”
- Here: juice for the day-1 mix and the day-3 feed **not yet tested**

Source: Wink (2008a, b), not peer-reviewed. Juice in this method is the note’s suggestion and has not been tried.

Drying a backup and reviving it

A pinch worked where a soaked spoonful failed own result

First attempt

A recipe worked out in a September 2026 conversation with Claude:

- a larger amount of flakes
- soaked in warm water first

It did not take.

October 2026

Flakes a year old:

- a good pinch, about 1 g, crushed into a fresh 30 g water + 30 g wholemeal/rye mix
- no separate soak

Clearly active in several days.

Why keep a backup: after a heat kill, it is the only way back to the original culture, rather than to whatever the flour supplies. That holds for the bacteria; the yeast may not come through drying.

Source: Own runs, September and October 2026; chapter 11, *Recovery Protocol*

Dry it cool: a wet film is heat-fragile, a dry flake is not

Drying it

- Spread thin on baking paper; room temperature until it snaps, about 24–48 h
- Crumble; store airtight, dry, cool and dark
- No oven, no warm dehydrator: the guide allows one only below 37°C

Dry is heat-proof; drying is not

- Until dry, the film is a wet culture: six hours or more at 40–45°C is high risk, and the D-value is under a minute at 55°C (chapter 11)
- Dried *S. cerevisiae* kept full viability through 60 min at 75°C, and 10 min at 100°C at 12% relative humidity
- The September chat said dried cells lose heat tolerance: backwards

In the kitchen: The heat risk is the wet start of drying, not the dry end: dry at room temperature.

Source: Country Roads Sourdough, not peer-reviewed; Ribert et al. (2021); chapter 11

Reviving it in five steps, with amounts by eye own result

1. **Start:** crush a good pinch of flakes, about 1 g, into about 30 g water + 30 g flour, half wholemeal, half rye. Water at room temperature or a little above, not cold; no separate soak
2. **Wait, unfed, for several days** at 24–26° C until clearly active. A dry crust forms on top: the day-2 skin, harmless
3. **Lift off the crust and feed what is underneath** with bread flour and wholemeal at about 1:1:1. From the next feed, keep 20 g and discard the rest, as on day 4
4. **Feed at peak** as in the main method; one or two feeds on the usual flour before baking with it
5. **Dry a new backup** once it doubles reliably

In the kitchen: Record flake age, rough amount, water temperature, first sour smell, first rise, first doubling. Sour without a rise: the LAB are back, the yeast not yet.

Source: The method as run here, October 2026; amounts estimated, not weighed, and it does not seem sensitive to them

A dried starter is not a paused starter

What	Through drying	Why
Water	Leaves, to equilibrium with the room	—
CO_2 and ethanol	Leave	A gas and a volatile, from a thin film
Acetic acid	About half leaves	Volatile: 57% kept in freeze-drying
Lactic acid	Mostly stays	Not volatile: 85% kept in freeze-drying
Spent flour: starch, gluten, bran, minerals	Stays	Already part-fermented: little food left in it
Flour enzymes	Stay, idle until wet	Flour is itself a dry product whose enzymes work once wetted
Cells	A minority survive dormant; most die	Next slide

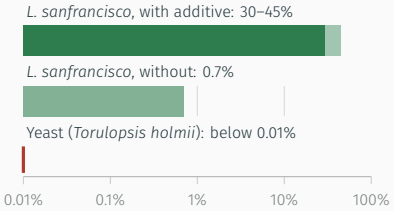
In the kitchen: Slow air-drying probably loses at least as much acetic acid, so a flake brings back mostly lactic acid. That is why the dose matters.

Source: Acid figures: Kline (1979), a patent, not peer-reviewed

Drying kills most cells, the yeast hardest

- First hours: the film keeps fermenting as sugars, salts and acids concentrate
  reasoning; nothing measured
- Damage: mechanical, structural and oxidative in yeast; in LAB mainly the membrane, as it changes phase
- Air-drying *S. cerevisiae*: under 1 in a million growing cells survive; 20–40% of cells that ran out of food
- **hypothesis** so dry later than the usual “at peak”? Uncertain: tolerance comes with respiration, a jar is short of oxygen, no sourdough yeast was tested (open test 4)

Survival, freeze-dried San Francisco starter



Kline (1979) **a patent, not peer-reviewed**

Freeze-drying is not air-drying, but this is the only sourdough-specific comparison found.

Source: Calahan et al. (2011); Dupont et al. (2014); Santivarangkna et al. (2008); Kline (1979), a patent, not peer-reviewed

Water starts four things at once

Rehydration

1. **The lactic acid redissolves.** A pinch gives the fresh flour a small dose; a spoonful gives close to full-strength starter acid

2. **Flour enzymes start freeing sugar again**, in the fresh flour and any left in the flake

3. **Survivors take up water** and their membranes pass back through the phase change: the moment of leakage, worse in cold water

4. **Dead cells leak** their contents into the mix

Then a lag, while the survivors repair before they divide. From there the revival is a race between their regrowth and the flour's own microbes.

In the kitchen: Step 3 is where water temperature matters: the leakage is worse in cold water.

Source: Santivarangkna et al. (2008); Crowe et al. (1989)

A pinch skips the first wave; the yeast still has to build up

From a pinch of dried flakes own result

Start: about 1 g crushed flakes + 30 g water + 30 g wholemeal/rye

First hours: lactic acid redissolves · flour enzymes free sugar · survivors rehydrate and repair before dividing

Surviving LAB regrow: many survivors, already acid-tolerant · pH falls below 4.5 → the flour's gassy first wave is shut out: no false rise, no lull

Yeast: perhaps a hundred survivors, maybe none, + newcomers from flour, hands, kitchen · the acid admits the tolerant ones

Several days, unfed: sour but no rise · dry crust on top · yeast regrows over about 20 doublings

First rise: lift off the crust · feed bread flour + wholemeal 1:1:1 · then keep 20 g, as on day 4

Feed at peak until it doubles reliably → dry a new backup

From scratch, for comparison

Days 1–4: the flour's first wave, *Leuconostoc*, *Enterobacter* · big rise, bad smell

skipped

Day 5: the lull · acid stops the first wave

Days 6–9: sourdough LAB take over · yeast the day after pH 3.5

Days 9–14: ratios climb until it triples

Source: own run, October 2026, for the timings. The order of events inside them is reasoning **no pH or counts were taken**

The surviving bacteria close the window the first wave needs

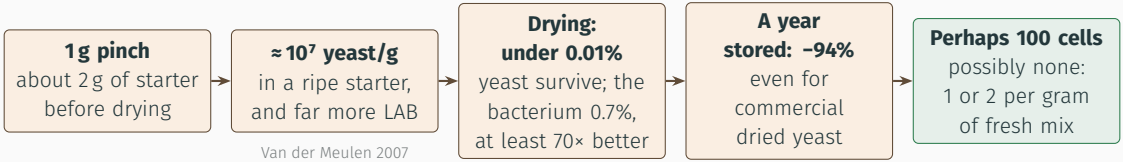
- From scratch, a gassy first wave rises and collapses and the jar goes quiet before the sourdough organisms take over
- The revival took days, but had **no false start and no lull** **own result**
- The likely reason is the inoculum: the surviving LAB are sourdough organisms, selected by months of feeding, that grow well at their own acidity
- They acidify the fresh flour before its first wave builds, closing Wink's window: the gas producers grow at pH ≥ 5 and not at ≤ 4.5
- From scratch nothing acid-tolerant is there in numbers, so the first wave gets days

reasoning that fits the run here; no pH was measured

In the kitchen: From flakes, expect quiet days that turn sour before rising: no day-4 surge, no lull.

Source: Wink (2008a), not peer-reviewed; own run, October 2026

The yeast restarts from perhaps a hundred cells, possibly none



- That is over 20 doublings short of a ripe starter: each tenfold is about 3.3 doublings
- The LAB start from far more survivors. The revival behaved as if the LAB had come through and the yeast had to be caught again **own result**
- **arithmetic on borrowed figures, not a count**: Kline's starter was freeze-dried, and Ribert's yeast was grown to be dried

Source: Van der Meulen et al. (2007); Kline (1979), a patent, not peer-reviewed; Ribert et al. (2021)

It skipped the bacterial half of the first week, not the yeast half

- **The yeast may not be the original.** Survivors compete with yeast arriving as it does from scratch: with the flour, and from the baker's hands
- A kitchen that kept the starter for years may hold its yeast: in four bakeries, the sourdough's dominant yeast also dominated walls, air, storage boxes and mixers. A "new" yeast may be the old one coming back **loose analogue**
- The yeast does not arrive in a from-scratch jar: the LAB have already acidified it. From scratch, yeast appeared the day after pH 3.5, so its clock starts at once, and the acid favours tolerant yeasts

In the kitchen: A backup reliably brings back the bacteria, and perhaps the yeast. Drying later may save more of it (open test 4); whether it is the old yeast is open test 7.

Source: Reese et al. (2020); Minervini et al. (2015): bakeries hold far more flour and dough than a home, and the study cannot say which way the yeast moved; Wink (2008a), not peer-reviewed

A pinch, not a spoonful: the flakes carry the old starter's acid

A spoonful
in a little water
and flour

↓

Close to full-strength starter acid, little fresh food: chapter 11's dying culture

first attempt, also warm-soaked: did not take **own result**

A pinch
in 60 g fresh wholemeal and rye

↓

Far more dilute, plenty of food; wholegrain is what chapter 11 adds

worked **own result**

- Flakes are the old flour plus all its lactic acid, which does not evaporate in drying
- Chapter 11 dilutes a dying culture out at 1:10:10
- Country Roads' recipe is the high-dose kind, 5 g of flakes into 20 g water + 20 g flour, and expects about a week
- Two things changed, amount and soak: one run cannot separate them (open test 6). The soak is the less likely culprit: 35–38°C for half an hour is brief by chapter 11's standards, unless the water was hotter

Source: Own runs, 2026; chapter 11, *Recovery Protocol*; Country Roads Sourdough, not peer-reviewed

Water: not cold. The documented hazard is cold, not warm

- The September chat advised lukewarm, 25–30°C, because cells are most fragile as water rushes back in. The fragility is real; the direction is wrong
- As a dry cell takes up water, its membrane lipids pass from gel to fluid. In cold water the membrane leaks while it changes state: **imbibitional damage**, measured in pollen, with dry yeast named as a case
- Active dry yeast rehydrated at 0°C against 40°C: less respiration and fermentation, membranes that did not reseal even after hours at 40°C, few cells growing on plates
- Limits: pollen; pure commercial *S. cerevisiae*; flakes in flour may wet more slowly; nothing about 20°C. The direction is supported; the room-temperature penalty is unknown (test 5)

In the kitchen: Mix with water at room temperature or a little above, up to about 35°C, then hold the jar at 24–26°C. Avoid cold tap water.

Source: Crowe et al. (1989); Peña et al. (1992); Country Roads Sourdough (room-temperature water; cold only slows things), not peer-reviewed

A backup loses ground every month, but slowly

Source	What was stored	Loss
Calahan et al. (2011)	Air-dried *S. cerevisiae*	About 20% survival over a month
Ribert et al. (2021)	Dried *S. cerevisiae*, 25°C, 55% relative humidity	1.24 log, about 94%, in a year
Hajar-Azhari et al. (2024)	Spray-dried, kombucha-started sourdough powder, best of four packagings **loose analogue**	About 2.5 log of LAB and yeast in 90 days
Here **own result**	Flakes a year old (October 2026)	Back from a pinch in several days

- A dry flake follows the room: water activity about 0.5 in a 50% kitchen, far below anything that grows. Losses are mainly oxidation of membrane lipids: hence airtight, cool, dark
- Each tenfold loss costs about 3.3 doublings: an old backup wakes more slowly, not never. True of the LAB; the yeast is less certain

In the kitchen: Replace the backup once a year from the live starter to keep the loss small.

Source: As in the table; FDA (1984); Santivarangkna et al. (2008); Country Roads Sourdough, not peer-reviewed; chapter 11

Takeaways

- The day-4 rise is bacteria and the day-5 lull is normal: keep to the schedule through both
- Feed at peak. When it falls before the next feed, go one ratio up
- Done means tripling in 4–6 h on 1:2:2 at 27°C. No hot builds before that
- Start warm, not hot: 24°C or 30°C is a choice of culture as well as pace
- The microbes come from the flour and your hands, not the air
- Dry a backup thin at room temperature; store it airtight, cool and dark; replace it yearly
- Revive with a pinch in a fresh wholemeal and rye mix, water not cold. The bacteria come back first; the yeast later, and perhaps a new one

Open tests

Seven open tests not yet tested

Starting from scratch

1. **Start temperature.** Two jars, same flour and day-1 mix: room temperature against 30°C (the chamber can hold it). Compare the end of the lull, the first tripling at 1:2:2, the smell. Van der Meulen et al. (2011) predicts different dominant LAB
2. **Does a 30°C start take to the hot protocol better** than a 24°C one? Only once both pass the tripling test, on split lineages (chapter 08)
3. **Juice versus water**, Wink's own design: two identical jars, one with juice on days 1–3

The dried backup

4. **Dry at peak or after it.** Two films from one jar, at peak and a few hours after the fall; revive equal weights; compare the first doubling. Calahan et al. (2011) predicts the later film wins
5. **Water temperature.** Same flakes, 35°C water against cold tap, both then at 24–26°C. Peña et al. (1992) predicts the cold jar is slower
6. **Pinch versus spoonful**: a pinch against 10 g, same flour and water temperature. The acid account predicts a slower spoonful; a pH strip at the start should show why
7. **Same yeast or a new one?** No kitchen test; sequencing the fungal ITS region of flakes and revived starter, as Reese et al. (2020) did, would. Dead cells keep their DNA; a species match would not prove the same strain

Source: The Open tests section of notes/starter-from-scratch.md

Sources

The method and the first two weeks

Kristen, Full Proof Baking (2019). *How To Make A Sourdough Starter From Scratch.* YouTube, 7 Feb 2019.

Wink D (2008a). The pineapple juice solution, Part 1. *Bread Lines* 16(1). Not peer-reviewed.

Wink D (2008b). The pineapple juice solution, Part 2. *Bread Lines*, June 2008. Not peer-reviewed.

Van der Meulen R et al. (2007). *Appl. Environ. Microbiol.* 73(15):4741–4750.

Van der Meulen R et al. (2011). *Appl. Environ. Microbiol.* 77(8):2716–2726.

Ercolini D et al. (2013). Microbial ecology dynamics during rye and wheat sourdough preparation. *Appl. Environ. Microbiol.* 79(24):7827–7836.

Minervini F et al. (2018). *Frontiers in Microbiology* 9:1984.

Reese AT et al. (2020). Influences of ingredients and bakers on the bacteria and fungi in sourdough starters and bread. *mSphere* 5(1):e00950-19.

Minervini F et al. (2015). House microbiotas as sources of LAB and yeasts in traditional Italian sourdoughs. *Food Microbiology* 52:66–76.

Drying and reviving

Country Roads Sourdough. *Dehydrating and Rehydrating Sourdough Starter: A How To Guide.* Not peer-reviewed.

Kline L (1979). Freeze-dried natural sour dough starter. US Patent 4,140,800. A patent, not peer-reviewed.

Calahan D et al. (2011). Genetic analysis of desiccation tolerance in *S. cerevisiae*. *Genetics* 189(2):507–519.

Dupont S et al. (2014). Survival kit of *S. cerevisiae* for anhydrobiosis. *Appl. Microbiol. Biotechnol.* 98(21):8821–8834.

Santivarangkna C et al. (2008). Inactivation mechanisms of LAB starter cultures preserved by drying. *J. Appl. Microbiol.* 105(1):1–13.

Crowe JH et al. (1989). Membrane phase transitions are responsible for imbibitional damage in dry pollen. *PNAS* 86(2):520–523.

Peña A et al. (1992). Rehydration temperature is critical for metabolic competence and membrane integrity in active dry yeast. *Arch. Microbiol.* 158(2):75–80.

Ribert P et al. (2021). Effect of devitrification on the survival and resistance of dried *S. cerevisiae*. *Appl. Microbiol. Biotechnol.* 105(16–17):6409–6418.

Hajar-Azhari S et al. (2024). Sourdough powder: shelf-life stability in different package type. *J. Food Sci. Technol.* 61(9):1701–1710.

US FDA (1984). *Water Activity (aw) in Foods.* Inspection Technical Guide.

Source: Full entries in the References section of notes/starter-from-scratch.md; chapter 11 of the main document